The Midnight Conservatory

WORKED SOLUTION KEY

01 Life Processes

MODEL ANSWERS • Match the number to the student assignment.

01 CBSE 2023 · 31/1/1 · Q8

It secretes digestive enzymes onto the bread. These break complex substances into simpler soluble molecules, which the fungus absorbs.

02 CBSE 2023 · 31/1/1 · Q9

Absorption decreases. The villi normally increase the surface area, and their capillaries carry absorbed nutrients away efficiently.

03 CBSE 2023 · 31/1/1 · Q23a

Body tissues → vena cava → right atrium → right ventricle → pulmonary artery → lungs. The blood picks up oxygen in the lungs.

04 CBSE 2023 · 31/1/1 · Q23b

Glucose first forms pyruvate in the cytoplasm. With enough oxygen, further breakdown produces CO_2 and water and releases more energy. Oxygen-deficient muscles convert pyruvate to lactic acid.

05 CBSE 2023 · 31/1/1 · Q24

Nephrons are in the kidneys. A capillary cluster filters small molecules from blood into Bowman's capsule. The tubule selectively reabsorbs useful substances and some water.

06 CBSE 2023 · 31/1/1 · Q29a

Respiration transfers energy to ATP. Breaking off its terminal phosphate releases usable energy, for example to power active transport across membranes.

07 CBSE 2023 · 31/1/1 · Q29b

They open their stomata at night, take in CO_2 and store it in organic acids. They release and use the CO_2 during the day while the stomata are mostly closed.

08 CBSE 2024 · 31/1/1 · Q11

The guard cells lose turgor and become less curved. The stomatal pore narrows or closes, reducing gas exchange and water loss.

09 CBSE 2024 · 31/1/1 · Q22

It provides the acidic conditions needed by pepsin and kills many microbes taken in with food.

10 CBSE 2024 · 31/1/1 · Q29

Thin walls shorten the diffusion path. Capillaries carry oxygen away and deliver CO_2, maintaining concentration gradients for exchange.

11 CBSE 2024 · 31/1/1 · Q29

Cartilage keeps the airway from collapsing. Residual air prevents the lungs from emptying completely and allows gas exchange between breaths.

12 CBSE 2025 · 31/1/1 · Q8

Salivary amylase starts breaking starch into smaller sugars such as maltose. Those products account for the sweeter taste.

13 CBSE 2025 · 31/1/1 · Q23a

Platelets help form a plug and initiate clotting. Limiting blood loss helps preserve blood volume and blood pressure, maintaining circulation to tissues.

14 CBSE 2025 · 31/1/1 · Q23b

Phloem moves dissolved sugars from sources, such as mature leaves, to sinks such as storage roots and growing shoots. Different sinks can lie above or below the source.

15 CBSE 2025 · 31/1/1 · Q38

$6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$ in light with chlorophyll. Twelve CO_2 molecules are twice the equation amount, giving 2 glucose and 12 O_2 molecules.

16 CBSE 2025 · 31/1/1 · Q38

The shaded plant receives less light energy. The blocked stomata restrict carbon dioxide entry, even though light is available.

17 CBSE 2026 · 31/1/1 · Q1

CO_2 enters through the stomata. A pair of guard cells controls each stomatal pore.

18 CBSE 2026 · 31/1/1 · Q5

Trypsin digests proteins. Lipase digests fats into fatty acids and glycerol. Bile emulsifies large fat globules, increasing the area available to lipase.

19 CBSE 2026 · 31/1/1 · Q9

Oxygen has limited solubility in plasma. Haemoglobin in red blood cells binds oxygen in the lungs and releases it in tissues, greatly increasing carrying capacity.

20 CBSE 2026 · 31/1/1 · Q10

The diaphragm flattens and increases the volume of the chest cavity. Lung pressure falls below atmospheric pressure, so air enters.

02 Control and Coordination

21 CBSE 2023 · 31/1/1 · Q11

It is positive hydrotropism. The direction of growth depends on the direction of the water stimulus.

22 CBSE 2023 · 31/1/1 · Q11

The root bends down and shows positive gravitropism. The shoot bends up and shows negative gravitropism.

23 CBSE 2023 · 31/1/1 · Q22

The pancreas releases insulin. It helps cells take up glucose and promotes storage as glycogen, lowering blood glucose towards its normal level.

24 CBSE 2023 · 31/1/1 · Q22

The stimulus for insulin release decreases, so the pancreas secretes less insulin. This negative feedback helps avoid a continuing fall in blood glucose.

25 CBSE 2024 · 31/1/1 · Q10

Arrival of the electrical impulse triggers release of a neurotransmitter. It crosses the gap and binds receptors on the next cell, where it may start a new electrical signal.

26 CBSE 2024 · 31/1/1 · Q23a

Mimosa folds rapidly through water-driven turgor changes, without growth. Root bending is growth-dependent and directed by gravity.

27 CBSE 2024 · 31/1/1 · Q23b

Thyroxine, secreted by the thyroid gland, regulates metabolism. Iodine is needed to synthesise it.

28 CBSE 2024 · 31/1/1 · Q23b

Insufficient iodine limits thyroxine production. Continued stimulation of the thyroid can enlarge the gland, producing a goitre.

29 CBSE 2024 · 31/1/1 · Q30

Skin receptor → Sensory neuron → Spinal cord → Motor neuron → Arm muscle.
The muscle contracts and pulls the hand away quickly.

30 CBSE 2024 · 31/1/1 · Q30

The spinal cord can initiate withdrawal through the reflex arc. Signals also reach the brain for conscious perception, but the rapid response can begin first.

31 CBSE 2025 · 31/1/1 · Q9

Auxin accumulates on the shaded right side of the shoot. Cells there elongate more, bending the shoot towards the light on the left.

32 CBSE 2025 · 31/1/1 · Q9

There is no stronger light stimulus on one side. Growth is more symmetrical, so little or no sideways phototropic bending is expected.

33 CBSE 2025 · 31/1/1 · Q29

The touch triggers signalling between cells. Water redistribution changes turgor in cells at the leaflet base, changing their shape and folding the leaf.

34 CBSE 2025 · 31/1/1 · Q29

Growth becomes unequal: the side away from contact elongates more than the contacted side. The tendril curves around the support in thigmotropism.

35 CBSE 2026 · 31/1/1 · Q4

The cerebellum coordinates precise voluntary movements and helps maintain posture and balance.

36 CBSE 2026 · 31/1/1 · Q4

The cerebrum is involved in planning and initiating voluntary action. The cerebellum coordinates the precision and timing of the movements.

37 CBSE 2026 · 31/1/1 · Q8

It is fast and automatic. The stimulus triggers a pre-existing nervous pathway without needing a conscious decision.

38 CBSE 2026 · 31/1/1 · Q11a

Starting to chew can be initiated voluntarily. Salivation in response to seeing food is an involuntary reflex.

39 CBSE 2026 · 31/1/1 · Q14

A sensory neuron carries information from the receptor to the central nervous system. A motor neuron carries a command from the central nervous system to the muscle.

40 CBSE 2026 · 31/1/1 · Q14

A neuron carries and processes information in the nervous system. A nephron is a kidney unit in the excretory system that filters blood and modifies the filtrate.

03 How Do Organisms Reproduce?

41 CBSE 2022 · 31/1/1 · Q3

Mustard and hibiscus flowers have both stamens and pistils. A pistillate papaya flower has no anthers of its own, so its pollen must come from a staminate flower.

42 CBSE 2022 · 31/1/1 · Q4

Spirogyra filaments break into fragments, each of which grows into a filament. Planaria can rebuild missing body parts through specialised cells so suitable pieces form complete animals.

43 CBSE 2022 · 31/1/1 · Q10

Testes form sperm and secrete testosterone. The scrotum helps maintain a temperature lower than core body temperature. The vas deferens carries sperm away from the testis region.

44 CBSE 2022 · 31/1/1 · Q10

Sperm are motile and have relatively little cytoplasm. The ovum lacks active swimming and contains more cytoplasm and stored materials for early development.

45 CBSE 2023 · 31/1/1 · Q12

A potato tuber is a modified stem. Its eyes are buds that grow into shoots. Bryophyllum plantlets arise from buds along the leaf margins.

46 CBSE 2023 · 31/1/1 · Q35

The ovary usually develops into a fruit and its wall becomes the fruit wall. The fertilised ovules develop into seeds.

47 CBSE 2023 · 31/1/1 · Q35

Sepals protect the developing bud. Petals can attract pollinators. Stamens and carpels contain the structures directly involved in reproduction.

48 CBSE 2024 · 31/1/1 · Q16

Plasmodium can reproduce by multiple fission. One parent produces several daughter cells, whereas binary fission produces two.

49 CBSE 2024 · 31/1/1 · Q35a

An intrauterine device is placed inside the uterus and interferes with fertilisation or related reproductive processes, depending on the type. Hormonal pills commonly prevent ovulation.

50 CBSE 2024 · 31/1/1 · Q35b

Spores can remain dormant on a dry surface. They need enough moisture and suitable conditions to germinate and grow into new mould.

51 CBSE 2025 · 31/1/1 · Q22

Repeated cell division produces an outgrowth. It enlarges and develops a mouth and tentacles. The mature bud detaches and lives independently.

52 CBSE 2025 · 31/1/1 · Q35a

The ovary releases ova. The oviduct carries the ovum and is the usual site of fertilisation. The uterus supports implantation and development.

53 CBSE 2025 · 31/1/1 · Q35a

A condom forms a barrier to semen entering the partner’s body and reduces transmission of many STIs. Vasectomy interrupts the vas deferens so sperm do not enter the semen.

54 CBSE 2025 · 31/1/1 · Q35b

Self-pollination uses pollen from the same flower or plant. Cross-pollination uses pollen from a different plant of the same species and can combine alleles from two plants.

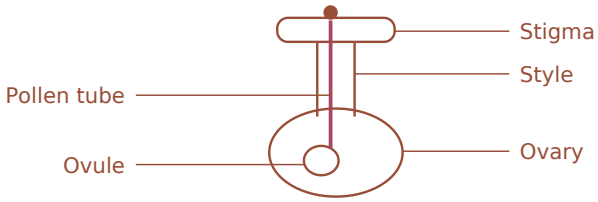
55 CBSE 2026 · 31/1/1 · Q2

Binary fission. The genetic material is copied and separated, then the cell divides into two daughter cells.

56 CBSE 2026 · 31/1/1 · Q11b

Pollination is pollen transfer to a stigma. Fertilisation still requires compatible pollen to germinate, a tube to grow and the male gamete to fuse with the egg.

57 CBSE 2026 · 31/1/1 · Q12



The stigma receives pollen. Its tube grows through the style and enters an ovule inside the ovary.

58 CBSE 2026 · 31/1/1 · Q16a

Fertilisation forms a zygote. The zygote divides as it moves towards the uterus. The developing embryo can implant in the uterine lining.

59 CBSE 2026 · 31/1/1 · Q16a

Hormone support for the lining falls. The lining breaks down and is shed with blood during menstruation.

60 CBSE 2026 · 31/1/1 · Q16b

The sporangium releases many spores. Air can carry them to new locations, where suitable moisture and nutrients allow them to germinate.

04 Heredity

61 CBSE 2022 · 31/1/1 · Q5b

Use V for violet and v for white. VV × vv gives all Vv. Selfing Vv yields VV, Vv, Vv, vv, so ¼ are white. Expected number = 160/4 = 40.

62 CBSE 2022 · 31/1/1 · Q5b

Yes. Violet plants can be VV or Vv. A Vv plant carries the recessive white allele, which is expressed only in vv plants.

63 CBSE 2022 · 31/1/1 · Q14

A typical body cell has 44 autosomes and two sex chromosomes. An ovum has 22 autosomes and one sex chromosome, X.

64 CBSE 2022 · 31/1/1 · Q14

X ovum + X sperm → XX.
X ovum + Y sperm → XY.
The sperm can contribute either an X or a Y chromosome.

65 CBSE 2022 · 31/1/1 · Q14

In some reptiles, the temperature at which eggs incubate influences sex. In the usual human model, it depends on the sex chromosomes inherited at fertilisation.

66 CBSE 2023 · 31/1/1 · Q10

A gene is a stretch of DNA carrying information for a functional product. A chromosome is a packaged DNA molecule with proteins and contains many genes.

67 CBSE 2023 · 31/1/1 · Q18

The ovum provides X, so the X chromosome comes from the mother. The Y-bearing sperm contributes Y.

68 CBSE 2023 · 31/1/1 · Q38

The two parents give RY and ry respectively. All F_1 are RrYy and have round yellow seeds because R and Y are dominant.

69 CBSE 2023 · 31/1/1 · Q38

Each gamete receives one allele from each pair. The combinations are RY, Ry, rY and ry, each expected with probability ¼.

70 CBSE 2023 · 31/1/1 · Q38

Wrinkled green requires rryy. The chance of rr is ¼ and that of yy is ¼. Multiply to obtain 1/16.

71 CBSE 2024 · 31/1/1 · Q8

It is established at fertilisation. An X-bearing ovum fuses with either an X-bearing sperm to form XX or a Y-bearing sperm to form XY.

72 CBSE 2024 · 31/1/1 · Q9

Meiosis halves the chromosome number to 23 per gamete. Fertilisation combines two sets, giving 23 + 23 = 46 in the zygote.

73 CBSE 2024 · 31/1/1 · Q38

Cross it with a short tt plant. TT × tt produces all tall Tt. Tt × tt gives an expected 1:1 ratio of tall Tt to short tt. A short offspring shows that the tall parent was Tt.

74 CBSE 2025 · 31/1/1 · Q11

The gametes are Ry and rY, giving RrYy. F_1 are round yellow: they show the dominant shape inherited from one parent and the dominant colour from the other.

75 CBSE 2025 · 31/1/1 · Q30a

Chromosomal DNA contains genes. Gametes carry parental copies of chromosomes. The resulting offspring inherits versions of genes that can influence its characteristics.

76 CBSE 2025 · 31/1/1 · Q30b

If gametes kept the full chromosome set, fertilisation would add two full sets. With the same process repeated, the chromosome number would double each generation.

77 CBSE 2026 · 31/1/1 · Q6

For one child, the probability is ½. With independence, multiply ½ × ½ = ¼, or 25%.

78 CBSE 2026 · 31/1/1 · Q15a

Each parent contributes T or t. The four combinations are TT, Tt, Tt and tt. Genotypes are 1:2:1. Phenotypes are 3 tall:1 short.

79 CBSE 2026 · 31/1/1 · Q15a

TT, Tt, Tt are the three equally likely tall outcomes in the Punnett grid. Two are heterozygous, so the fraction among tall offspring is ⅔.

80 CBSE 2026 · 31/1/1 · Q16a

Heat-tolerant bacteria are more likely to survive and reproduce in warmer water. Their descendants inherit the variant, so its frequency can rise.

05 Our Environment

81 **CBSE 2022 · 31/1/1 · Q7a**

Ultraviolet light splits some O_2 into oxygen atoms. A free O atom can combine with O_2 to form O_3. The ozone layer absorbs much of the harmful ultraviolet radiation.

82 **CBSE 2022 · 31/1/1 · Q7b**

A pond has a larger, more diverse food web and nutrient cycles. In an aquarium, wastes can accumulate quickly and food, oxygen, filtration and water quality may need human management.

83 **CBSE 2022 · 31/1/1 · Q13**

Dead matter and waste would accumulate. Nutrients would be returned to soil and water more slowly, reducing their availability to producers.

84 **CBSE 2023 · 31/1/1 · Q26**

Small organisms take up the pesticide and retain it. Fish eat many contaminated organisms, and birds eat many fish. The concentration can rise at successive trophic levels through biomagnification.

85 **CBSE 2023 · 31/1/1 · Q33a**

Aquatic plants or algae use light to produce food and oxygen. Microbes break down waste and recycle nutrients. A balanced stocking level helps prevent oxygen depletion and excessive waste.

86 CBSE 2023 · 31/1/1 · Q33b

Snakes receive about 0.10 × 20,000 = 2,000 J. Hawks receive about 0.10 × 2,000 = 200 J.

87 CBSE 2024 · 31/1/1 · Q15

A primary consumer is missing. A grass-eating mouse provides a plausible link: grass → mouse → snake → hawk.

88 CBSE 2024 · 31/1/1 · Q18

The relevant mechanism is persistence and retention. Predators repeatedly consume contaminated prey, which can increase the pesticide concentration along the chain.

89 CBSE 2024 · 31/1/1 · Q33a

Plants may increase as grazing decreases, and lions lose a food source. A food web can buffer the effect if lions switch to other available prey.

90 CBSE 2024 · 31/1/1 · Q33b

Chlorofluorocarbons, or CFCs, release ozone-destroying chlorine in the stratosphere. The Montreal Protocol controls production and consumption of many ozone-depleting substances.

91 CBSE 2025 · 31/1/1 · Q15

Snakes do not eat grass and frogs are not typical predators of snakes. An example is grass → grasshopper → frog → snake.

92 CBSE 2025 · 31/1/1 · Q16

Captured energy is 0.01 × 80,000 = 800 J. The remaining 80,000 − 800 = 79,200 J is not stored as food.

93 CBSE 2025 · 31/1/1 · Q20

Formation and destruction are different processes. The historical decline is explained by catalytic ozone destruction by substances such as chlorine released from CFCs.

94 CBSE 2025 · 31/1/1 · Q26

Runoff carries pesticide into pond water, altering its chemical quality. Aquatic organisms can be poisoned directly or exposed through contaminated food.

95 CBSE 2026 · 31/1/1 · Q7

Banana peel and ordinary paper can be decomposed by microbes. Polythene and a conventional plastic pen barrel persist and are treated as non-biodegradable.

96 CBSE 2026 · 31/1/1 · Q13

Primary consumers receive about 4,000 J. Secondary consumers receive about 400 J. Tertiary consumers receive about 40 J.

97 CBSE 2026 · 31/1/1 · Q13



Each successive level receives less usable energy.

98 CBSE 2026 · 31/1/1 · Q13

One step back gives primary consumers 120/0.10 = 1,200 J. Another gives producers 1,200/0.10 = 12,000 J.

99 CBSE 2026 · 31/1/1 · Q14

A consumer obtains food from other organisms. Decomposers such as many bacteria and fungi break down dead material and release nutrients into the environment.

100 CBSE 2023 · 31/1/1 · Q33b

The secondary consumer level needs 25/0.10 = 250 J. The primary consumer level needs 250/0.10 = 2,500 J.

The reference shelf

Each citation identifies the source idea in a CBSE Class 10 Science board paper. Prompts are paraphrased, split into individual skills, or extended with fresh data. Question labels refer to the source paper, and worksheet numbers refer to this assignment.

2022 · Science 086 · Paper 31/1/1 · Term II

Open question paper reference

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